

1 Convergence in $\mathbb{R}$ - (Euclidean topology)

Definition. Let $\{x_n\}_n$ be a sequence in $\mathbb{R}$ and $x \in \mathbb{R}$. We say that $x_n \rightarrow x$ (as $n \rightarrow \infty$) if

$$\forall \epsilon > 0, \exists N \in \mathbb{N} \text{ s.t. } |x_n - x| < \epsilon, \forall n \geq N$$

Resolving the absolute value in the definition above we can rewrite the definition of convergence as follows

$$\forall \epsilon > 0, \exists N \in \mathbb{N} \text{ s.t. } x - \epsilon < x_n < x + \epsilon, \forall n \geq N$$

which intuitively says that points of the sequence x_n lie in the interval $]x - \epsilon, x + \epsilon[$ for n large enough.

2 Continuity in $\mathbb{R}$ - (Euclidean topology)

Definition. Let $A \subseteq \mathbb{R}$ and $f : A \rightarrow \mathbb{R}$ be a function.

- We say that f is continuous at $x \in A$ if

$$\forall \epsilon > 0, \exists \delta > 0 \text{ s.t. } |f(y) - f(x)| < \epsilon, \forall y \in A \text{ s.t. } |y - x| < \delta.$$

- We say that f is continuous on A if f is continuous at x , for all $x \in A$.
- We will use the following notation

$$C(A) = \{f : A \rightarrow \mathbb{R} \mid f \text{ is continuous on } A\}$$

Examples

- All polynomial functions are continuous on $\mathbb{R}$
- Each rational function (ratio of two polynomials) is continuous on $\mathbb{R} \setminus Z$ where Z is the zero set of its denominator.
- Exponential functions (that is $x \mapsto a^x$ where $a > 0$) are continuous on $\mathbb{R}$
- the set of continuous functions $C(A)$ is closed under addition, subtraction, and multiplication. This means that the sum, difference, and product of continuous functions (on A) results in a continuous function (on A).
- The ratio of two continuous functions on A is a continuous function on $A \setminus Z$ where Z is the zero set of the denominator function.
- The composition of continuous functions is a continuous function.

3 Open sets in $\mathbb{R}$ - (Euclidean topology)

Definition. Let $A \subseteq \mathbb{R}$.

- We say that x is an interior point of A if there exists $r > 0$ such that $]x - r, x + r[\subseteq A$.
- We say that A is open if all of its point are interior points, that is

$$\forall x \in A, \exists r > 0 \text{ s.t. }]x - r, x + r[\subseteq A$$

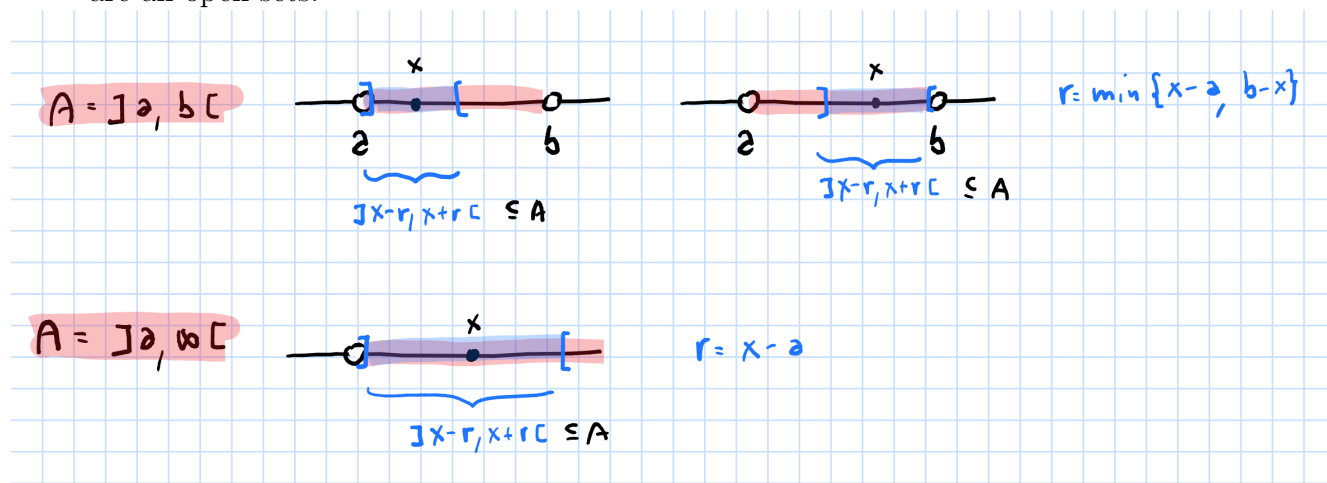
So, a set A is open if every point of A is contained in an 'open' interval that is contained in A . Intuitively, if you start at a point in A , you can walk a little bit to right or to the left without leaving the set A .

Examples

- The empty set is open. (It vacuously satisfies the definition).
- All intervals of the form $]a, b[$ where $-\infty \leq a < b \leq \infty$ are open sets. (This is because if you take a point $x \in]a, b[$ you can consider the radius r equal to the minimum distance of x to the endpoints of the interval). For example,

$$A_1 =]0, 1[\quad A_2 =]-\sqrt{3}, \frac{\pi}{e}[\quad A_3 =]5, \infty[\quad A_4 =]-\infty, 1[\quad \mathbb{R} =]-\infty, \infty[$$

are all open sets.



- Unions of intervals of the form above are open sets. For example

$$A_1 =]-1, 3[\cup]7, 10[\quad A_2 =]-\infty, 0[\cup]0, 3[\cup]100, \infty[\quad A_3 = \bigcup_{n \in \mathbb{N}}]n, n + \frac{1}{3}[$$

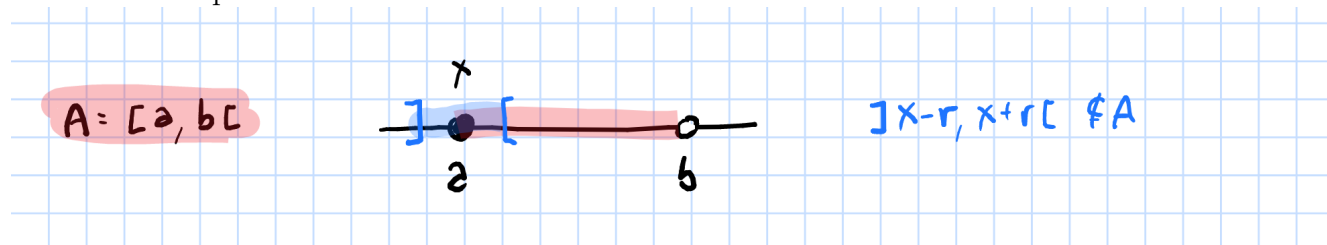
are all open sets.

Nonexamples

- i) Intervals of the form $[a, b[$ where $-\infty < a < b \leq \infty$ are not open. (This is because $a \in [a, b[$ but no interval of the form $]a - r, a + r[$ with $r > 0$ is contained in $[a, b[$). For example,

$$A_1 = [-1, 2[\quad A_2 = [4, \infty[$$

are not open.



- ii) Similarly, intervals of the form $]a, b]$ where $-\infty \leq a < b < \infty$ are not open. For example,

$$A_1 =]-1, 2] \quad A_2 =]\infty, 4]$$

are not open.

- iii) Singletons $\{a\}$ where $a \in \mathbb{R}$ are not open,

4 Exercises

Exercise 1 Determine which of the following sets are open in $\mathbb{R}$ (with the Euclidean topology)

- i) $]0, 1[\cup \{2\}$
- ii) $[a, b]$ where $a, b \in \mathbb{R}, a < b$
- iii) $\mathbb{R} \setminus \{0\}$
- iv) $\bigcup_{n \in \mathbb{N}} [n^{-1}, 5 - n^{-1}]$